UNIVERSITY OF RWANDA

College of Science and Technology

Visit Rwanda

Mobile Application — UI/UX Design Report

A Flutter-based Travel Discovery & Booking App for Rwanda

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Course	Mobile Application Systems and Design
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1. Project Overview

Visit Rwanda is a multi-screen Flutter mobile application designed to help tourists and travelers discover, explore, and book travel experiences across Rwanda's most iconic destinations. The app features a clean, nature-inspired green color palette reflecting Rwanda's identity as the Land of a Thousand Hills.

The application consists of three primary screens:

- Home Screen — browse and search destinations with category filters
- Detail Screen — explore rich destination information, gallery, and reviews
- Booking Screen — complete a travel reservation with dynamic pricing

The app is built using Flutter and Dart, following a clean widget-based architecture with reusable components, responsive layout, and a consistent design language throughout.

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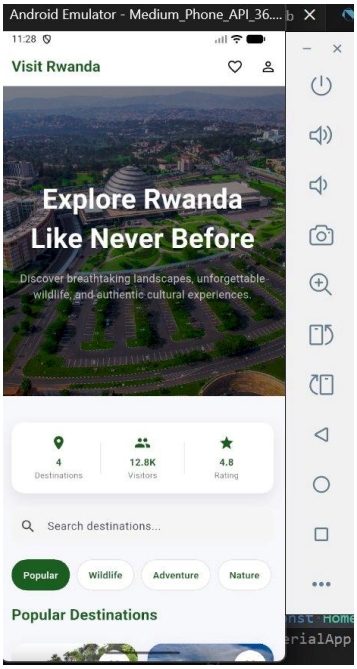
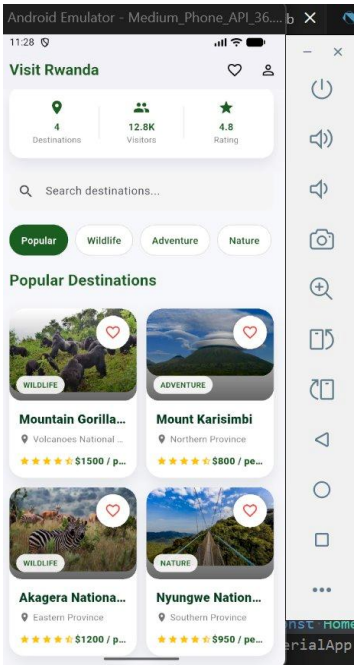
2. Screen Screenshots

2.1 Home Screen



Home Screen — Hero & Stats

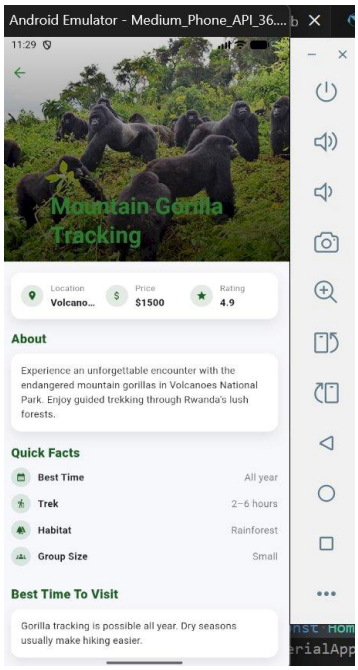


Home Screen — Destination Grid

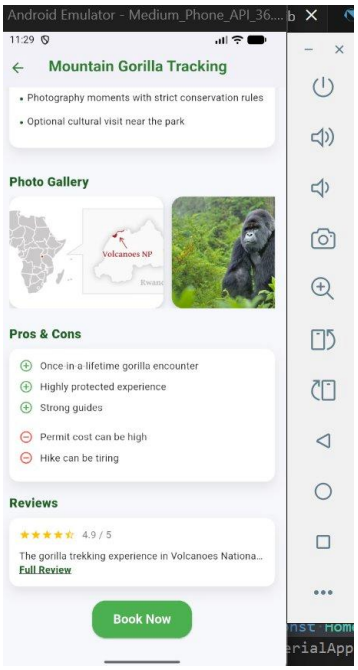
The Home Screen is the main entry point of the app. It opens with a full-bleed hero image of Rwanda overlaid with bold introductory text and a semi-transparent dark scrim for readability.

Below it, a statistics card summarises key metrics. A search bar and horizontal category filter chips allow the user to browse destinations filtered into a responsive 2-column grid of destination cards.

2.2 Detail Screen



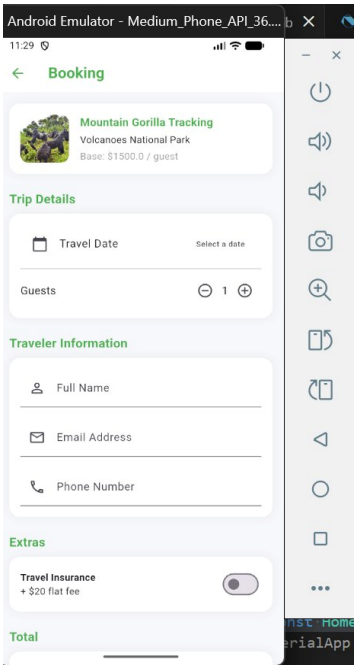
Detail Screen — Info & Facts



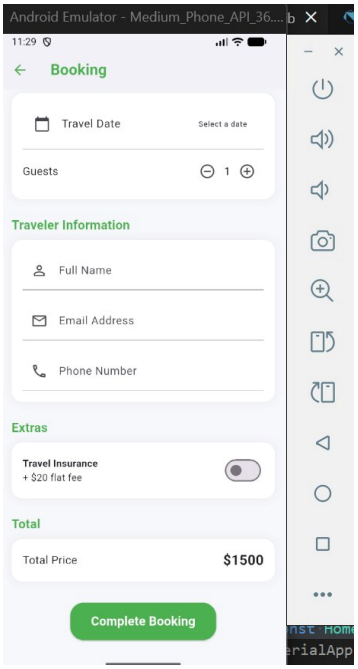
Detail Screen — Gallery & Reviews

The Detail Screen uses a SliverAppBar with a collapsible hero image. As the user scrolls, the app bar collapses to a pinned green title. Content is organised into clearly labelled sections inside white card containers: About, Quick Facts, Best Time to Visit, Activities, Photo Gallery, Pros & Cons, Reviews, and a Book Now CTA button.

2.3 Booking Screen



Booking Screen — Form Top



Booking Screen — Total & CTA

The Booking Screen presents a clean form-driven layout. A destination summary card reminds the user which experience they are booking. Sections below collect Trip Details (date picker, guest counter), Traveler Information (name, email, phone), and Extras (travel insurance toggle). A dynamically computed Total Price updates in real time. Confirming shows a SnackBar and pops back to the Home Screen.

3. Layout & Design Choices

3.1 Color Palette

The primary accent color is deep forest green (#1B5E20), chosen to represent Rwanda's lush natural environment and national identity. White card backgrounds provide clean contrast for content, while light grey (#F5F7FA) is used as the scaffold background to add subtle depth between sections. Red is reserved exclusively for the favorite heart icon, signaling interactivity without cluttering the green theme.

3.2 Typography

Bold 40sp headlines in the hero section establish strong visual hierarchy immediately. Section titles use 22sp bold green text to guide the reading flow. Body copy uses 16sp at a 1.5 line-height for comfortable mobile reading. Card titles use 18sp weight-900 for strong emphasis. All text uses Flutter's default Roboto font family which provides excellent legibility on mobile screens.

3.3 Hero Section

A full-width 450px-tall Container with a BoxDecoration image and a 45% opacity black overlay ensures text remains legible over any background photo. Center-aligned white text creates a cinematic, immersive first impression consistent with travel app conventions.

3.4 Card-based Layout

Every major content block on the Detail and Booking screens is wrapped in a white card with rounded corners (borderRadius: 14–18) and a soft drop shadow (blurRadius: 14–18, offset: Offset(0, 8)). This floating-card UI pattern is widely recognised in modern mobile design and helps users visually group related information at a glance.

3.5 Responsive Grid

The destination grid on the Home Screen uses MediaQuery to switch between a 2-column layout (screens below 520px) and a 3-column layout (wider screens). This makes the app functional on small phones as well as tablets without separate layout code.

3.6 Sticky Collapsible AppBar

The Detail Screen uses a CustomScrollView with SliverAppBar (expandedHeight: 300, pinned: true). This pattern lets the hero image expand on load and gracefully collapse into a compact title bar as the user scrolls, maximising content real estate while maintaining navigation context.

3.7 Spacing & Padding

Consistent 16px horizontal padding is used throughout all screens. Vertical spacing between sections uses SizedBox(height: 18–24) for comfortable breathing room. Cards use 14–18px internal padding. This systematic use of whitespace creates a calm, uncluttered feel appropriate for a premium travel app.

4. Widgets Used

Widget	Screen(s)	Purpose
AppBar	Home, Booking	Top navigation bar with title and action icons
SliverAppBar	Detail	Collapsible hero bar with pinned title on scroll
Scaffold	All screens	Base layout structure providing body and app bar
SingleChildScrollView	Home, Booking	Scrollable column layout for long-form content
CustomScrollView	Detail	Sliver-based scroll for collapsing hero header
GridView.builder	Home	Responsive 2-column destination card grid
ListView.builder	Home, Detail	Horizontal scrolling for chips and photo gallery
TextField	Home, Booking	Search input and traveler information form fields
Container	All screens	Custom decoration boxes — hero, cards, category chips
Stack + Positioned	Home, Detail	Overlay elements: gradient, category tag, heart icon
Image.asset	All screens	Local asset images for destinations and hero section
ClipRRect	Card, Booking	Rounded corner image clipping on cards
InkWell	Card, Detail	Tap ripple effect on destination cards and gallery
IconButton	All screens	Favorite toggle, back arrow, profile icon actions
ElevatedButton	Detail, Booking	Primary CTAs: Book Now and Complete Booking
Switch	Booking	Travel insurance on/off toggle
ListTile	Booking	Travel date picker trigger row with trailing text
CircleAvatar	Detail	Circular icon badge in info tiles and quick facts
FlexibleSpaceBar	Detail	Flexible content area inside SliverAppBar
GestureDetector	Detail	Tap handler for full review dialog and gallery zoom

Widget	Screen(s)	Purpose
AlertDialog	Detail	Full review modal popup with scrollable text
InteractiveViewer	Detail	Pinch-to-zoom gallery image fullscreen viewer
SnackBar	Booking	Booking confirmation feedback notification
DatePicker (showDatePicker)	Booking	Native OS date selection dialog
MediaQuery	Home	Screen width detection for responsive grid columns
CategoryChip (custom)	Home	Pill-shaped filter chip: selected vs unselected state
DestinationCard (custom)	Home	Destination preview card: image, rating, price, heart
RatingStars (custom)	Home, Detail	Star rating row with half-star support (0–5 scale)
PrimaryButton (custom)	—	Reusable full-width elevated button component

5. Navigation Flow

The app uses Flutter's imperative Navigator with MaterialPageRoute push/pop transitions. There are no named routes — each screen is pushed directly with the destination model passed as a constructor argument, keeping data flow explicit and type-safe.

From	Action	To	Data Passed
App Launch	main() → runApp()	Home Screen	None
Home Screen	Tap destination card	Detail Screen	Destination model
Detail Screen	Tap Book Now button	Booking Screen	Destination model
Booking Screen	Tap Complete Booking	Home Screen (pop)	SnackBar feedback
Detail Screen	Tap back arrow	Home Screen (pop)	None
Booking Screen	Tap back arrow	Detail Screen (pop)	None
Detail Screen	Tap Full Review text	AlertDialog (overlay)	Review text string
Detail Screen	Tap gallery thumbnail	Full-screen dialog	Image asset path
Booking Screen	Tap Travel Date field	showDatePicker	DateTime result

5.1 Navigation Flow Diagram

The diagram below maps every screen transition and overlay trigger in the Visit Rwanda application. Green nodes represent full screens; purple nodes represent modal overlays and dialogs. Solid arrows indicate forward navigation pushes; dashed arrows indicate back-navigation pops or dismissals.

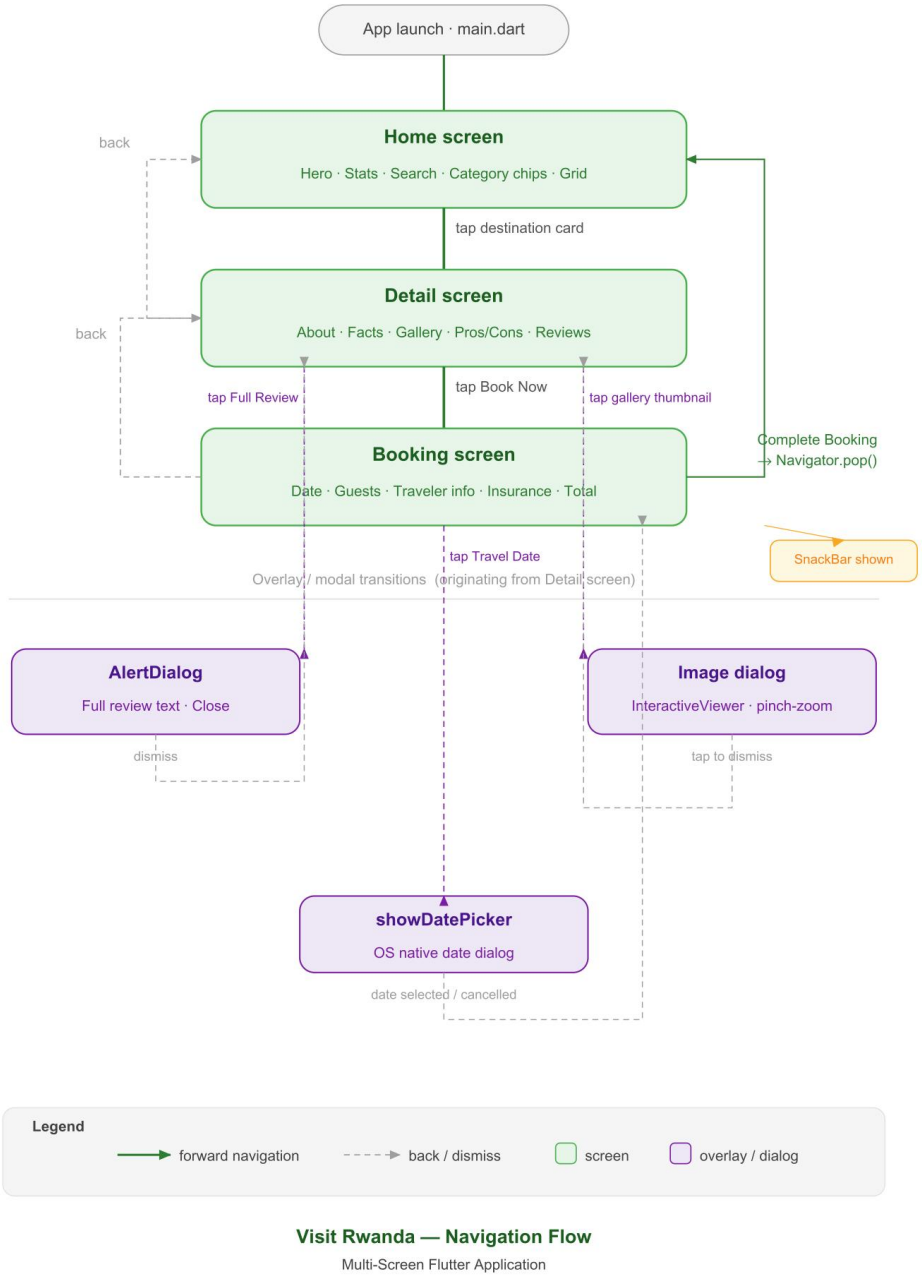


Figure 5.1 — Visit Rwanda full navigation flow diagram

6. Project Architecture

The project follows a clean folder-based architecture separating concerns into distinct layers:

Folder / File	Responsibility
lib/main.dart	App entry point, MaterialApp theme configuration
lib/models/destination_model.dart	Destination data class (id, title, location, price, rating, category, image, description)
lib/data/destinations.dart	Static data — destination list and category list
lib/screens/home_screen.dart	Home Screen: stateful widget, search, filter, grid logic
lib/screens/detail_screen.dart	Detail Screen: stateless widget with all content sections
lib/screens/booking_screen.dart	Booking Screen: stateful widget, form state, price logic
lib/widgets/destination_card.dart	Reusable card: image, category tag, heart, rating, price
lib/widgets/category_chip.dart	Reusable filter chip with selected/unselected state styling
lib/widgets/rating_widget.dart	Star rating row with half-star rendering (RatingStars)
lib/widgets/custom_button.dart	Reusable full-width PrimaryButton component
assets/images/	Local destination images, detail gallery assets, maps

7. Conclusion

The Visit Rwanda application demonstrates a complete, production-quality Flutter mobile app following modern UI/UX principles. The app provides an end-to-end user journey from discovery to booking, with careful attention to visual hierarchy, responsive layout, consistent branding, and intuitive navigation. The reusable widget architecture ensures maintainability and scalability for future additions such as user authentication, live backend integration, payment processing, and map-based destination browsing.